

Fluency Drill Book

Use these short drills for warm-ups, stations, tutoring, homework, or re-teach cycles.

Recommended timing: 6 minutes for attempts, 4 minutes for correction and strategy talk.

Part 1: Decimals and Estimation

Drill 1A: Compare and Round

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|----------------------------------|--------------------------------------|---------------------|
| 1. 4.06 ____ 4.6 | 5. Write 7.082 expanded. | 9. $0.48 \div 10$ |
| 2. 0.305 ____ 0.350 | 6. Order: 2.2 , 2.02 , 2.202 . | 10. $0.48 \div 100$ |
| 3. Round 18.947 to tenths. | 7. 10×0.48 | |
| 4. Round 18.947 to hundredths. | 8. 100×0.48 | |

Drill 1B: Decimal Operations

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|---------------------|-------------------------------|------------------|
| 1. $6.4 + 2.87$ | 5. $15.6 \div 3$ | 9. $2.35 + 0.8$ |
| 2. $12 - 3.58$ | 6. $8.4 \div 0.7$ | 10. $5.01 - 0.9$ |
| 3. 0.7×0.6 | 7. Estimate 39.8×5.1 | |
| 4. 4.5×0.2 | 8. Estimate $81.2 \div 9.8$ | |

Part 2: Fractions

Drill 2A: Equivalence

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|-------------------------------------|--|--|
| 1. $\frac{3}{5} = ?/10$ | 5. $\frac{5}{12}$ ____ $\frac{1}{2}$ | 8. Find a common denominator for $\frac{2}{5}$ and $\frac{1}{3}$. |
| 2. $\frac{4}{6}$ simplest form | 6. Write a fraction equal to $\frac{9}{12}$. | 9. Place $\frac{7}{10}$ near 0, $\frac{1}{2}$, or 1. |
| 3. $\frac{6}{8}$ simplest form | 7. Find a common denominator for $\frac{1}{4}$ and $\frac{1}{6}$. | 10. Place $\frac{11}{12}$ near 0, $\frac{1}{2}$, or 1. |
| 4. $\frac{2}{3}$ ____ $\frac{3}{4}$ | | |

Drill 2B: Operations

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|---------------------------------|-------------------------------------|-----------------------------------|
| 1. $\frac{1}{4} + \frac{1}{8}$ | 5. $\frac{2}{3} \times 9$ | 9. $1\frac{1}{2} + 2\frac{1}{4}$ |
| 2. $\frac{2}{3} + \frac{1}{6}$ | 6. $\frac{3}{4} \times \frac{2}{5}$ | 10. $4\frac{1}{3} - 2\frac{2}{3}$ |
| 3. $\frac{5}{6} - \frac{1}{3}$ | 7. $5 \div \frac{1}{5}$ | |
| 4. $\frac{3}{5} - \frac{1}{10}$ | 8. $\frac{3}{4} \div 3$ | |

Part 3: Ratios, Rates, Percent

Drill 3A: Ratios and Rates

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|-------------------------------------|---|--|
| 1. Simplify 6:9. | 5. 2 cups to 5 cups scaled by 4 = ____ to ____. | 8. 45 pages in 5 min = ____ pages/min. |
| 2. Simplify 12:16. | 6. Are 4:6 and 10:15 equivalent? | 9. Complete: 7:4 = 21:____. |
| 3. 3 for \$12 = \$____ each. | 7. 8 pencils for \$2.40 = \$____ each. | 10. Complete: 5:8 = ____:40. |
| 4. 150 miles in 3 hours = ____ mph. | | |

Drill 3B: Percent Benchmarks

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|--------------|----------------------|--------------------------|
| 1. 10% of 90 | 5. 20% of 35 | 9. \$24 with 25% off |
| 2. 50% of 76 | 6. 15% of 80 | 10. Estimate 49% of 102. |
| 3. 25% of 48 | 7. 75% of 40 | |
| 4. 1% of 600 | 8. \$30 with 10% off | |

Part 4: Algebra and Coordinates

Drill 4A: Expressions

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|------------------------------------|-------------------------------------|-------------------|
| 1. Evaluate $3x + 2$ when $x=5$. | 4. Write: twice the sum of a and 4. | 7. 4 squared + 3 |
| 2. Evaluate $20 - 4n$ when $n=3$. | 5. $8 + 2 \times 6$ | 8. $18 / 3 + 7$ |
| 3. Write: 6 less than y. | 6. $(8 + 2) \times 6$ | 9. $5(2 + 6)$ |
| | | 10. $30 - 12 / 4$ |

Drill 4B: Equations and Points

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|-----------------|--|--|
| 1. $x + 9 = 20$ | 5. Check $x=11$ in $x+9=20$. | 8. Distance from (2,1) to (2,7). |
| 2. $m - 7 = 18$ | 6. Plot (3, 6): x or y first? | 9. Write three points with $x=4$. |
| 3. $6p = 54$ | 7. Same y-coordinate means horizontal or vertical? | 10. Write an equation: a number times 3 is 27. |
| 4. $q / 8 = 5$ | | |

Part 5: Geometry and Data

Drill 5A: Measurement

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|----------------------------------|----------------------------------|---------------------------------------|
| 1. Area: 9 by 7 | 5. Surface area of cube side 3 | 9. Which unit: pencil length? |
| 2. Perimeter: 9 by 7 rectangle | 6. Missing side: area 56, side 8 | 10. Which unit: classroom floor area? |
| 3. Triangle area: $b=12$, $h=5$ | 7. Convert 3 ft to inches. | |
| 4. Volume: $4 \times 3 \times 8$ | 8. Convert 2 yd to feet. | |

Drill 5B: Data

1. Mean: 2,4,6,8
2. Median: 9,3,7,5,5
3. Mode: 1,2,2,3,4
4. Range: 11,15,20,22
5. Outlier in 4,5,6,7,50
6. Best graph for categories?
7. Best graph for change over time?
8. Add one value to make mean of 4,6,8,? equal 7.
9. Find total if mean is 9 for 5 values.
10. Explain median in one sentence.

Part 6: Mixed Readiness

Drill 6A: Multi-Step

1. 6 packs of 8 minus 13
2. $\$7.50 \times 4 + \6
3. $15 \times 8 + 3 \times 4$
4. $\frac{3}{4}$ of 20
5. 25% of 72
6. $2.4 \times 5 + 1.6$
7. $x + 14 = 39$
8. 5 pens cost \$6. What is 1 pen?
9. Area of triangle $b=9$, $h=4$
10. Estimate 198×49

Drill 6B: Explain and Audit

1. Find the mistake: $\frac{1}{2} + \frac{1}{3} = \frac{2}{5}$.
2. Find the mistake: $0.4 \times 0.5 = 2.0$.
3. Which strategy solves $4n=36$?
4. Which strategy solves a recipe ratio?
5. Write an estimate for 51×19 .
6. What unit for volume?
7. What does the denominator tell?
8. What does "per" mean?
9. Why check with substitution?
10. Write one question before solving a word problem.

Answer Key

Drill	Answers
1A	<; <; 18.9; 18.95; $7 + 0.08 + 0.002$; 2.02, 2.2, 2.202; 4.8; 48; 0.048; 0.0048.
1B	9.27; 8.42; 0.42; 0.9; 5.2; 12; about 200; about 8; 3.15; 4.11.

2A	6; $\frac{2}{3}$; $\frac{3}{4}$; <; <; $\frac{3}{4}$ or equivalent; 12; 15; near $\frac{1}{2}$ or 1; near 1.
2B	$\frac{3}{8}$; $\frac{5}{6}$; $\frac{1}{2}$; $\frac{1}{2}$; 6; $\frac{3}{10}$; 25; $\frac{1}{4}$; $3\frac{3}{4}$; $1\frac{2}{3}$.
3A- 6B	Use teacher guide for selected keys; accept equivalent ratios, equivalent fractions, and correct explanations.